

The smooth-germ programme: a synthesis (germbij Theorem 3.1’s analytic bridge, five stages)

Tide loop (Claude Fable 5, with GPT-5.6 Sol deliberation)

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Abstract

Five stages (PRs #60–#63) carried the germbij note’s Theorem 3.1 from its polynomial algebraic core to the smooth setting: two admissible C^{R+2} losses whose normalized moments at x^{2+r} differ by $o(q^{2+2r})$ for every $r \leq R$ have the same Taylor data through degree $R+2$ (**smooth_jet_recovery**), with nondegeneracy $\lambda > 0$ derived from the admissibility envelope rather than assumed. The programme’s central design constraint — Taylor replacement produces order-controlled errors, never exact data — shaped every interface: recovery theorems with pure **Tendsto** hypotheses, an epsilon-radius jet form, and a comparison theorem whose split radius $\sqrt{q}$ keeps **rpow** out of the entire development.

1 What was proven

- **StableRecovery** (PR #60): the recovery interface with all data hypotheses as vanishing limits at the coefficient-sensitive scales — eleven lines of composition, because the weighted-jet programme’s internal theorems were already limit-based.
- **TaylorPackage** (PR #61): the stabilizer envelope (coefficient-wise d , case split at $|x| = \rho$, no suprema), exact Gaussian scaling in q -space, and the envelope tail bound with the superpolynomial prefactor.
- **TaylorCompare** (PR #62): the admissibility package (partition lower bound $2\delta e^{-\kappa\delta^2}q$, moment bound $C_s q^{s+1}$) and the programme’s hardest step — the local Taylor comparison, unnormalized at $o(q^{s+D-1})$ and normalized at $o(q^{s+D-2})$, by the $\sqrt{q}$ split, the exponential secant bound with the Taylor power kept inside the Gaussian, and a **Metric.tendsto_nhds** epsilon-of-room assembly whose quantifier order matches the theorem’s.
- **SmoothRecovery** (PR #63): the Taylor adapter — the Peano remainder from Mathlib, the derived minimum facts, the Taylor polynomial in jet shape, and the stabilized jet’s joint certificate of admissibility and profile — plus the headline composition.

2 Design decisions that carried the programme

Three consult-driven choices did disproportionate work. The *uniformly positive profile* envelope (from the weighted-jet programme) survived unchanged into the smooth setting: the stabilized Taylor polynomial inherits it by construction, with the stabilizer placed above the Taylor degree so its scaled contribution is invisible at every recovered rung. The $\sqrt{q}$ *split* made the outer region’s

superpolynomial factor $e^{-c/q}$ elementary and banished `Real.rpow` from five files of asymptotics. And the *limit-based interfaces* maintained at every layer meant the “asymptotically stable” variants the smooth setting requires were compositions, not reproofs.

3 The composition-time discovery

The one structural surprise arrived exactly where the original scoping consult pointed (“full-jet quantifier management”): the recovery induction quantifies over every rung of its coefficient vector, but the stabilized jet is deliberately longer than the data reaches. The honest statement is the induction truncated at rung R (`jet_recovery_stable_partial`); the full-length theorem is simply inapplicable, since the data genuinely says nothing about the stabilizer coefficient. Interface mismatches of this kind seem to be the characteristic failure mode of composition stages, worth budgeting for in future programme scoping.

4 Process

Five tides, each with the consult-or-prior-deliberation record, executed numerical checks (with one caught fabrication near-miss, documented in the C2 log, and the rule sharpened: the check runs in one command, the log text quotes its output in a later one), and the post-incident build gate (import line plus fresh `.olean`) verified at every checkpoint. The composition plan for the final stage was written into the tide log *before* the Lean, which let the assembly proceed mechanically across several hundred lines.

5 What remains

The germbij nondegenerate arc is now formalised end to end. The note’s open commissions: the multivariate Theorem 3.1 (the Susceptibility Primer’s missing Stage 3 — Wick machinery on $\mathbb{R}^d$, a foundations programme), and the full-jet quantifier packaging (C6), which would upgrade the finite-order theorem to equality of whole Taylor series. Both need fresh scoping consults.